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Breakthrough in Downhole Flow Profiling Using Advanced 3D Thermal-Hydrodynamic Modeling in EMRC Wells

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Abstract

Horizontal drilling has become the industry standard for oil and gas development due to its extended reservoir exposure and improved recovery efficiency. However, accurately evaluating flow behavior in horizontal completions remains a complex challenge, especially in long-lateral wells and environments hostile to conventional sensors. This paper focuses on the use of advanced diagnostic technologies, specifically, 3D thermal-hydrodynamic modeling and acoustic power spectrum logging, to evaluate flow profiles and completion integrity in Extended Maximum Reservoir Contact (EMRC) wells offshore Abu Dhabi. The objective is to present a robust solution for identifying inflow performance and assessing swell packer sealing in extreme horizontal environments.

Two EMRC wells, with laterals ranging from 15,000 to 25,000 feet, were selected for detailed downhole diagnostics. The completions consisted of limited-entry liners supported by oil-swell packers. To assess both inflow performance and packer sealing integrity, a combination of high-precision temperature logging, acoustic power spectrum tools, and an array-based production logging tool (PLT) was deployed. The acquired data were processed using a newly developed 3D temperature modeling simulator that models thermal and hydrodynamic flow in the near-wellbore region. The modelling operates on a spatial-temporal grid that includes the wellbore, near-reservoir environment, and historical production timelines, allowing detailed reconstruction of flow dynamics across the survey interval.

The diagnostics revealed successful zonal isolation in both wells in most places, confirming the integrity of the oil-swell packers. The 3D modeling engine provided a high-resolution flow profile that demonstrated excellent agreement with the array PLT measurements, validating the accuracy of the thermal-hydrodynamic modelling. Furthermore, the combination of temperature modeling and acoustic data provided clear insights into behind-casing flow behavior, something conventional tools cannot capture. The 3D hydrodynamic temperature modeling also resolved flow geometries at the toe, heel, and along fractures, giving a complete picture of reservoir contribution across the lateral.

Traditional production logging tools face multiple challenges in horizontal wells, particularly in Extended Maximum Reservoir Contact (EMRC) wells, where mechanical sensors often fail. In contrast, the

temperature and acoustic tools deployed in this field are resistant to debris, viscous effects, and mechanical degradation. The newly developed modeling engine introduces a breakthrough capability with its 3D grid and transient physics-based equations, offering flow quantification with unprecedented accuracy. This marks a significant step forward in smart diagnostics for complex completions and provides operators with a reliable method to enhance well performance evaluation in long horizontal wells.

Introduction

The global decline in conventional hydrocarbon reserves has driven the development of advanced extraction technologies to economically exploit reservoirs in hard-to-reach locations. Extended Maximum Reservoir Contact (EMRC) wells significantly enhance reservoir drainage by maximizing lateral exposure, enabling production from distant zones relative to the vertical drilling axis. While highly effective, these wells involve complex and costly wellbore trajectories, necessitating rigorous performance monitoring to justify their economic viability.

This study focuses on production surveillance in EMRC wells, which exhibit homogeneity, implying uniform productivity across zones. Production Logging Tools (PLT) remain a primary method for direct flow measurements due to their interpretative simplicity. However, multistage horizontal wells present unique challenges, including complex multiphase flow regimes, fluid segregation, and post-stimulation flowback. Modern PLT tools now integrate multiple spinners, resistivity, capacitance sensors, and optical probes to address these complexities.

Despite advancements, conventional PLT remains limited to evaluating flow inside the liner, providing no data on external contributions from fractures or matrix influx.

A Cost-Effective Alternative is High-Precision Temperature - Spectral Acoustic Logging. To overcome these limitations, an R&D initiative developed an innovative approach combining 3D thermo-hydrodynamic modeling with spectral acoustic data (DAS/DTS or, preferably, data from single-point sensors like High-Resolution Temperature and Acoustic Power Spectrum logging). This method enables:

- Comprehensive flow characterization (inside and outside liners, including perforated completions)
- Quantitative assessment of fracture vs. matrix contributions
- Accurate rate and phase fraction estimation

Key Advantages Over PLT

- Efficient fluid entry identification in long horizontals
- Early detection of water/gas breakthrough
- Reduced operational risks in EMRC wells
- Lower costs compared to full PLT suites

Field cases demonstrate strong agreement between PLT and High-Resolution Temperature and Acoustic Power Spectrum logging results. Where PLT data is unreliable (e.g., low rates, logging restrictions), temperature modeling supplements interpretation, ensuring robust reservoir performance analysis.

For EMRC operators, High-Resolution Temperature and Acoustic Power Spectrum logging offers a reliable, economical solution for routine well surveillance, bypassing the operational hurdles of traditional PLT while delivering critical insights into near-wellbore and far-field flow dynamics.

This paper details the thermo-hydrodynamic model, data acquisition methodology, and case studies validating the approach.

Theory and definitions

This study presents an innovative approach for detecting and quantifying fluid influxes in Extended Maximum Reservoir Contact (EMRC) horizontal wells by integrating advanced acoustic power spectrum analysis and high-resolution temperature logging with numerical modelling. The logging techniques deployed incorporate substantial enhancements over conventional methods, including upgraded hardware systems and refined data processing algorithms. Below is a concise overview of the methodologies employed.

Acoustic Power Spectrum

Fundamental Principles. The movement of fluids or gases through porous media or completion strings produces distinct acoustic noise signatures. These signals originate from complex physical interactions, including pore-scale fluid dynamics and elastic vibrations within the rock matrix. Although downhole noise logging is a mature technology, modern spectral noise logging—featuring high-frequency sampling at reduced intervals—now provides superior resolution for analyzing wellbore and reservoir flow behavior.

The Acoustic Power Spectrum system utilizes an integrated hydrophone to capture acoustic emissions, converting them into electrical signals that are amplified, digitized, and stored in downhole memory. This high-fidelity amplitude-versus-time dataset enables advanced post-processing.

A key constraint of noise logging arises from the underlying physics: detectable acoustic energy is governed by both flow velocity and the reservoir's pressure gradient. For instance, high-rate laminar flows through large conduits often generate noise below the acoustic tool's detection threshold due to minimal pressure differentials.

Acoustic Power Measurement Methodology. To minimize motion-induced artifacts, the Acoustic tool collects data at stationary intervals, with optimal acquisition times of 45-90 seconds per station in horizontal wells. Typical station spacing ranges from 6-10 ft but is adjustable based on Wellbore geometry, Formation properties, and Operational requirements.

Efficiency Enhancement. The synchronized daisy-chain sensor configuration allows tripling the station interval while preserving data resolution. This innovation reduces field operation time without compromising diagnostic capability, as all sensors are pre-calibrated and time synchronized.

Data Processing Workflow and Interpretation. The data processing begins with Fast Fourier Transformation (FFT) to convert time-domain acoustic measurements into frequency-domain power spectra covering the 0.117-60 kHz range for each monitoring station. Advanced proprietary algorithms then perform three critical functions: first, they separate statistically relevant flow signals from background noise; second, they enhance spectral contrast to improve interpretability; and third, they generate depth-correlated spectral profiles that maintain the spatial relationship of flow features.

The processed results are visualized as depth-indexed spectrograms where amplitude is color-coded - red indicates high amplitude signals, yellow represents moderate amplitudes, and blue shows low amplitude measurements. Frequency characteristics provide diagnostic information about flow paths, with higher frequency components typically corresponding to fluid movement through smaller conduits such as micropores, while lower frequencies suggest flow through larger features like fractures.

Classification of Acoustic Signatures. Four distinct noise categories can be identified through spectral analysis:

Wellbore flow generates characteristic signals in the 0-1 kHz range, primarily caused by turbulent flow vibrations in the annulus. These include gas breakout signatures below 5 kHz that exhibit a progressive downward frequency shift as they approach the surface. Unlike other noise types, wellbore flow signals maintain consistent amplitude and frequency characteristics across all depths.

Completion-related noise produces broadband emissions with depth-specific high-amplitude peaks. These originate from various well components, including perforation tunnels, packers, and potential casing leaks. Each completion element creates a unique acoustic fingerprint that can be identified through careful spectral analysis.

Behind-casing crossflow manifests as vertically confined spectral patterns whose characteristics are fracture-dependent. Variations in frequency distribution along these vertical features provide valuable insights into changes in fracture geometry and flow dynamics.

Reservoir flow signatures show distinct frequency ranges: pore-scale liquid flow typically generates 10-30 kHz noise, while gas movement through pores produces even higher frequencies. Fracture-associated flow appears in the 3-5 kHz range, which may sometimes overlap with wellbore noise. These reservoir flow characteristics demonstrate a strong correlation with formation permeability profiles, making them particularly valuable for reservoir evaluation.

Thermodynamic modeling

One of the most effective tools for obtaining quantitative interval-by-interval inflow or injectivity profiles (including behind-liner flow profiles) in horizontal wells is the 3D hydrodynamic temperature modeling software suite. This is an advanced temperature-based 3D hydrodynamic simulator designed to model complex flow geometries in the near-wellbore region under intricate completion configurations. The simulator is capable of incorporating multiple active fractures and accounts for the dynamic interaction between wellbore hydraulics and reservoir thermodynamics, making it especially suitable for reservoir characterization and production diagnostics in challenging environments.

General Physical Overview. This section outlines the core principles of a coupled thermo-mechanical model specifically designed for interpreting temperature logging data in horizontal wells intersecting multiple hydraulic or natural fractures. The model aims to simulate heat and fluid transport processes occurring within the complex well-reservoir system to improve diagnostic capabilities and assess inflow contributions along the wellbore.

The physical setup is illustrated in [Figure 1](#), which presents a schematic of a horizontal well completed within a single-porosity, anisotropic reservoir. The reservoir is assumed to be bounded above and below by impermeable formations, ensuring no vertical fluid exchange beyond the reservoir interval. The well trajectory is characterized using True Vertical Depth (TVD) plotted against measured depth (MD), typically showing a low-angle inclination due to the horizontal alignment of the wellbore through the productive interval.

The reservoir fluid is considered to be a homogeneous multiphase mixture, which may include oil, water, and high-pressure gas. For modeling purposes, this fluid mixture is treated as a single pseudo-fluid with effective thermophysical properties such as density, viscosity, specific heat capacity, and thermal conductivity, all of which may vary with pressure and temperature but are averaged for the domain.

The reservoir matrix is assumed to be anisotropic, meaning that the permeability and thermal conductivity can differ in the vertical and horizontal directions. However, the lateral variations in matrix properties are assumed to be smooth or slowly changing, allowing for numerical simplification without neglecting key heterogeneities.

Each fracture intersecting the horizontal well is explicitly modeled as a planar, rectangular feature. These fractures are defined by several key geometrical parameters:

- Half-length (L_f): The horizontal extension of the fracture on each side of the wellbore.
- Height (h_f): The vertical extent of the fracture within the reservoir.
- Vertical position (z_f): The depth of the fracture relative to the reservoir base.

- Aperture (5f): The fracture opening width, which affects both fluid flow and thermal exchange across the fracture face.

Together, these features allow the model to capture the complex interplay between fluid dynamics and heat transfer in fractured horizontal wells, enabling the interpretation of high-resolution temperature data and other thermal logs for advanced reservoir diagnostics and production optimization.

Various completion configurations—such as open hole sections, slotted or pre-packed screens, intervals isolated by casing packers in conjunction with inflow control devices (e.g., sliding sleeves or interval control valves in SMART completions), as well as perforated liners and plug-and-perf systems—can all be treated as specific cases or modifications of the generalized schematic depicted in [Figure 2](#).

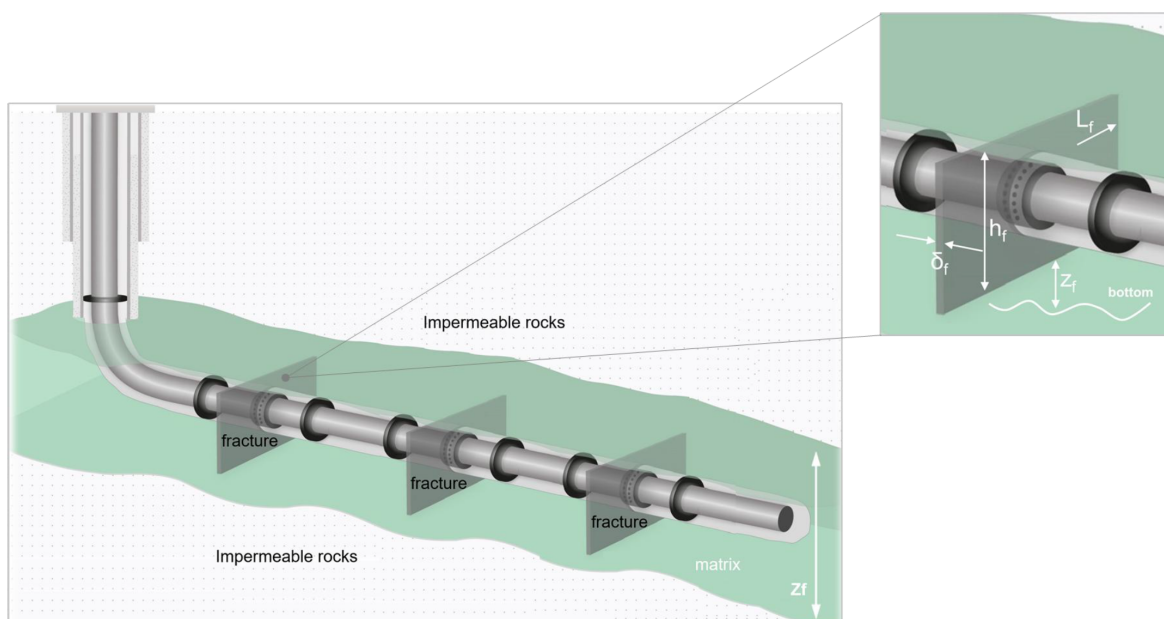


Figure 1—Schematic of the 3-D rock/reservoir model domain with a multi-fractured wellbore.

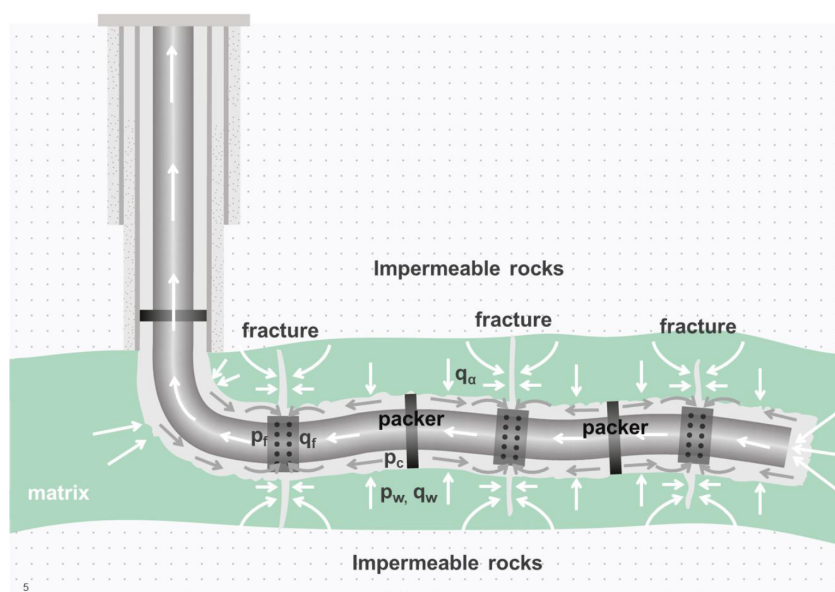


Figure 2—Detailed schematic of horizontal well components important for temperature modelling.

Reservoir hydrodynamic model. According to the modeling framework (Alekseeva et al., 2021), the reservoir hydrodynamics incorporates a predefined number of vertical transverse fractures, each characterized by distinct geometric and hydraulic properties. The unsteady three-dimensional fluid flow and pressure distribution within the porous reservoir matrix are governed by Darcy's law (Charny, 1963), formulated within a locally defined pseudo-Cartesian coordinate system. In this framework, the longitudinal **s-axis** aligns with the trajectory of the horizontal wellbore (see Fig. 1).

Within each fracture, vertical fluid flow and pressure are assumed to be laterally uniform and are computed under a quasi-steady-state approximation (Cinco et al., 1978). This allows for the characterization of inflow/outflow rates q_a and skin factors at the fracture-wellbore intersection points.

The overall hydraulic behavior of the SMART-completed well is captured by a system of equations describing multiphase fluid movement along the liner and across annular flow paths segmented by sand screens and mechanical packers. These governing equations are based on the work of Charny (1963), Altshul (1970), Bratland (2010), Ouyang et al. (1998), and Ozkan et al. (1993), and account for wellbore storage effects and additional pressure losses (skin effects) at the sandface and within the inflow control devices (e.g., sliding sleeves, perforations).

To simulate flow in the anisotropic matrix, the model utilizes an analytical integral approximation of the pressure field. This method treats the wellbore and fractures as time-dependent distributed sinks/sources (Carslaw and Jaeger, 1959; Tikhonov and Samarskii, 1966). The pressure in the flowing zones of the wellbore p_f and annular space p_c is calculated in tandem with the sandface pressure profile p_w , while simultaneously solving for inflow/outflow rates from fractures q_f and from the matrix q_w .

This hydrodynamic solver is capable of simulating various operational regimes of a horizontal well—including transient injection, production, and shut-in conditions—and can capture complex 3D flow behavior in the near-wellbore region and within the fracture network.

Temperature model. The convective and conductive heat transfer processes are described using a transient conjugate model that builds upon classical and modern thermophysical studies (Charny, 1961; Chekaliuk, 1965; Pudovkin et al., 1977; Aslanyan et al., 2013; Aslanyan et al., 2014). This comprehensive model accounts for dynamic temperature fields in the reservoir matrix and bounding impermeable formations, as well as within hydraulically active fractures and along the internal wellbore domains—including both liner flow and annular crossflows.

A key feature of this thermal transport engine is its inclusion of calorimetric mixing, as well as thermodynamic effects, such as adiabatic expansion/compression and Joule-Thomson cooling/heating, which collectively influence the effective hydrothermal conductivity of the system.

Horizontal wells with complex completions—such as those involving multiple hydraulic fractures and SMART control elements—represent physically intricate systems. Their thermohydraulic behavior is governed by multifaceted heat exchange interactions, which significantly shape the resulting temperature distributions along the wellbore. Accurate interpretation of temperature logging data, therefore, requires rigorous treatment of interrelated laminar, transient, and turbulent flow regimes, as well as transitions between forced, free, and mixed convection, particularly within the liner and annular domains. Foundational studies (Bratland, 2010; Mikheev and Mikheeva, 1973; Keis, 1972; Dzhaluriya, 1980) offer critical insights into these flow phenomena under varying thermohydrodynamic conditions.

A sophisticated sub-model for coupled heat exchange—formulated and demonstrated in detail by Virt et al. (2021)—has been integrated into the overall heat transfer framework. This model iteratively computes dynamic heat exchange coefficients, accounting for the thermal interaction between annular fluid streams and the sandface, as well as cross-flow heat transfer through liner walls between internal and annular streams.

The numerical implementation of the temperature simulation engine relies on finite-difference approximations over a structured computational grid (Samarskii, 1971; Paskonov et al., 1984), enabling stable and accurate resolution of transient thermal fields across all domains of interest.

Field overview and reservoir properties within the survey zone

A Comprehensive Overview of Reservoir

The field is located offshore Abu Dhabi, approximately 84 kilometers northwest of the capital island. This reservoir has been a cornerstone of the UAE's oil production since the field's discovery in 1963. It represents one of Abu Dhabi's most prolific carbonate reservoirs,

The reservoir contains a substantial hydrocarbon column of 250-300 feet, comprising a 32° API oil layer beneath a gas cap. Fluid analysis shows a gas-oil ratio of approximately 500 scf/bbl and a sulfur content of 1.8%. Average reservoir properties include 25% porosity (ranging from 18-28%) and 40 mD permeability (ranging from 30-300 mD), with oil saturation averaging 75%. These characteristics have supported impressive initial production rates of 8,000-12,000 barrels per day per well.

Development of the offshore field has progressed through several phases. Following initial 3D seismic campaigns between 2006-2015, operators drilled 15 appraisal wells with full core coverage to better understand reservoir characteristics. Since 2018, 4D seismic monitoring has been implemented to track fluid movement and optimize recovery. Current recovery factors stand at about 35%, with ambitious plans to reach 70% through enhanced oil recovery techniques. One of these techniques is drilling Extended Maximum Reservoir Contact wells.

Key challenges in developing the reservoir include managing the slightly tilted oil-water contact (which varies by about 15 meters across the field), addressing permeability contrasts between different shoal facies, and controlling gas cap expansion through careful production rate management. The reservoir's relatively homogeneous nature allows for more straightforward development, though carbonate-specific challenges like wettability effects and potential fracture networks require careful consideration.

The reservoir continues to play a vital role in Abu Dhabi's oil production strategy, with ongoing technological advancements ensuring its position as a key asset for decades to come.

Extended Maximum Reservoir Contact (EMRC) Wells: Technology and Applications

Extended Maximum Reservoir Contact (EMRC) wells represent a significant advancement in hydrocarbon recovery technology, pushing the boundaries of conventional horizontal well design. These engineered well configurations typically achieve between 20,000 to 30,000 feet of productive zone exposure - three to five times greater than standard horizontal wells - through sophisticated 3D well paths or multilateral architectures.

The completion design of EMRC wells incorporates several innovative features to manage the challenges of extended reach. Operators typically install 30 or more isolated zones using swellable packers, combined with a slotted liner to regulate fluid entry and future conformance issues.

From a reservoir engineering perspective, EMRC wells offer substantial advantages. The dramatically increased contact area improves recovery factors to 60-75%, compared to 30-45% for conventional wells, while simultaneously reducing the field development footprint. This technology proves particularly valuable in offshore environments where platform slot availability is constrained, or in mature fields requiring precise pressure maintenance.

However, the extended reach of EMRC wells introduces unique operational challenges. The exceptional length of productive intervals creates complex flow dynamics, including significant frictional pressure losses along the lateral and uneven fluid inflow distributions. These characteristics demand specialized

logging and monitoring solutions, often requiring tractor-assisted operations and memory logging tools to overcome deployment challenges in the extended laterals.

The formation development serves as an excellent case study for EMRC success. The operator has successfully implemented this technology to maximize recovery from the carbonate reservoirs while maintaining economic viability.

Equipment and operational procedures

Description of the integrated High-Precision Temperature - Acoustic Power Spectrum tool

The well evaluation program employed an integrated High-Precision Temperature - Acoustic Power Spectrum logging toolstring deployed on slickline conveyance. To facilitate navigation through the horizontal completion, a downhole tractor system was utilized, providing the necessary thrust force for tool advancement. The system operated in memory mode, with all acquired data being stored in downhole modules for subsequent surface retrieval and processing.

This configuration offered several operational advantages (Fig.3):

- Slickline deployment enabled efficient tool runs without requiring fluid circulation
- Downhole tractor assistance ensured reliable tool passage through lateral sections
- Memory-mode operation eliminated the need for real-time data transmission

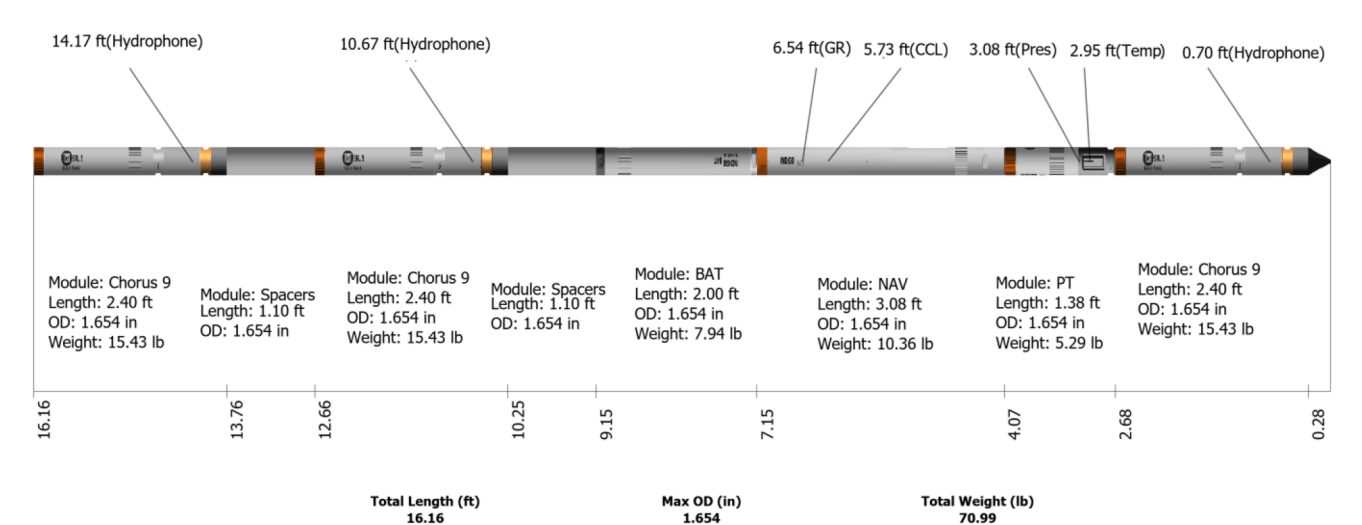


Figure 3—Tool String

The memory-mode recording system captured high-resolution datasets at programmed intervals throughout the logging run, with quality control parameters verified during surface data download. This approach proved particularly effective in the extended-reach horizontal environment, where conventional logging methods might encounter deployment challenges. The tractor-assisted slickline method provided optimal tool positioning while minimizing operational time compared to alternative conveyance methods.

The toolstring included the following sensors:

Table 2—Sensors that included the toolstring

Sensor	Mnemonics	Function
Casing collar locator	CCL	Depth correlation, completion position verification

Sensor	Mnemonics	Function
Gamma ray	GR	Referencing formation units, identification of radiogeochemical anomalies
Pressure gauge	PRES	Wellbore pressure measurement
High Precision Temperature Tool	TEMP	Temperature measurement
Acoustic Power Spectrum Tool	CHORUS	Identification of drainage zones, packer integrity

Problem Statement and Data Acquisition Plan

Drilling and Completion teams were using Oil swellable packer for Oil wells and Water Swellable packer for water injection wells. All extended lateral wells are horizontal and drilled with RDF-NAF (non-aqueous fluid) mud. The oil swellable packer disintegration was observed while deploying the liners in the horizontal section due to premature swelling with extended exposure to oil-based mud and friction encountered while running > 25,000 ft long liner string with 25-30 swell packers. Drilling and completion team switched to a slip-on type water swellable packer system for Oil wells as well to avoid premature swelling and disintegration of the Oil swellable packer; however, the slip-on type of swell packer was found to be susceptible to friction and started breaking off while the liner was lowered in the open hole.

Swell packer pieces were recovered at the shale shaker when oil-based mud was circulated out with brine, raising concern about the swell packer's ability to swell and isolate the compartment for effective stimulation and zone conformance. Since water swellable packers were installed for oil wells and the wells were producing >5% BS&W, the field development team wanted to assess the integrity of swell packers to ensure zonal conformance. (swell packer maintaining seal between liner and 8.5" open hole)

The study evaluated two wells to demonstrate the capabilities of the integrated Temperature - Acoustic toolstring in assessing swellable packer integrity and inflow profiles within Extended Maximum Reservoir Contact (EMRC) completions. The horizontal section, drilled through the oil-saturated formation, features an extensive 25,000 ft approximately reservoir drain completed with 31-35 zones isolated by swell packers. Lower completion type is Limited entry liner with Dissolvable plugged nozzles (LEL-DPN).

Swell Packer Details (See Fig.4):

- Swell Packers: Total 36qty water swellable, 3ft long.
- Elastomer: Water Swellable, 3ft long; running OD: 8.025".
- Type: Slip-on.
- Base-pipe: 6-5/8" 24ppf TSH W523 13Cr-80.
- Sealing ID / Maximum: 8.5" / 9.2".
- Pressure differential: 1000 psi.

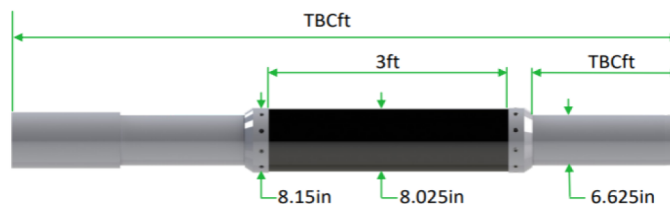


Figure 4—Swell Packer configuration

Logging Program

Due to continuous production requirements, the wells could not be shut in for logging operations. As the wells had been in sustained production before the survey, the acquisition program was designed to maintain flowing conditions throughout the operation. The logging procedure, consistent across both study wells (as illustrated in Figure 5), comprised two key phases:

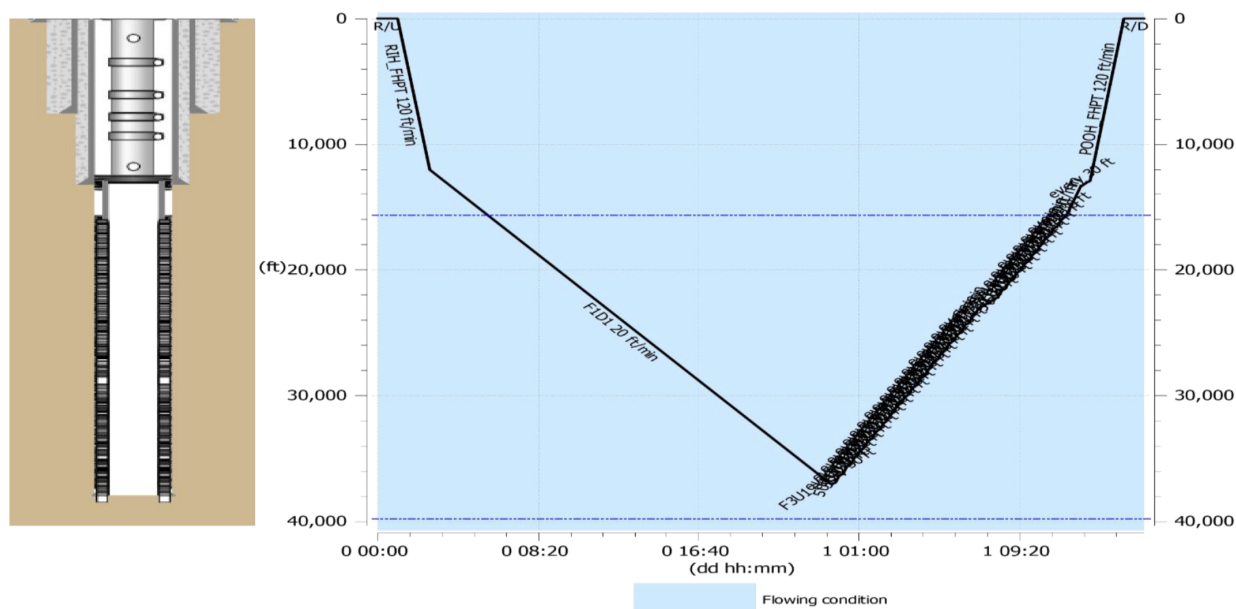


Figure 5—High Precision Temperature - Acoustic Power Spectrum survey schematic

Temperature Survey: Conducted during the initial downhole pass at a controlled speed of 20 ft/min to establish baseline thermal profiles under production conditions.

Acoustic Power Spectrum Logging: Performed during the uphole pass using stationary Acoustic power measurements spaced at 30 ft intervals supporting daisy chain technologies (three tools together), providing high-resolution acoustic data across the entire completion.

This sequential approach enabled comprehensive evaluation of both thermal characteristics and flow dynamics while maintaining well production. The combination of temperature and spectral acoustic signal data provided complementary datasets for robust interpretation of packer performance and inflow distribution along the extended-reach lateral.

Key Features of the Acquisition Methodology:

- Memory-mode operation ensured data integrity throughout the extended logging runs.
- Downhole tractor assistance facilitated tool navigation through the long horizontal section.
- Optimized station spacing balanced data resolution with operational efficiency.
- Flowing condition measurements provided representative well performance data.

The acquired datasets enabled detailed analysis of completion integrity and reservoir inflow patterns, as discussed in the following interpretation sections.

Case study #1

Well history

Extended Reach Well Survey and Analysis. The offshore horizontal producer Well #1 represents an advanced Extended Maximum Reservoir Contact (EMRC) completion in the carbonate formation. Drilled and completed in January 2019, the well features a 24,865 ft (7.58 km) lateral section — a remarkable length for single-bore horizontal completions in carbonate reservoirs at the time.

Completion Details. The 6.625" limited-entry liner completion incorporates swellable packers dividing the lateral into 35 discrete production zones. This design enables controlled inflow distribution along the extensive drain section while maintaining mechanical integrity. Notably, the well achieved its production targets without requiring hydraulic fracture stimulation, demonstrating the effectiveness of the extended contact approach in this reservoir.

Production History. After 32 months of continuous production, the well maintained a stable flow rate of 6,850 BOPD at the time of the July 2021 survey. The production stream showed minimal water cut or gas breakthrough, confirming effective sweep efficiency across the lengthy completion. Bullhead stimulation was conducted within a year since put on production.

Advanced Diagnostic Survey. The comprehensive July 2021 evaluation employed cutting-edge downhole monitoring technology, High-precision temperature logging, and Acoustic power spectrum tools.

These tools successfully characterized 23 of the 35 completion zones, providing crucial insights into:

- Zonal production contributions
- Swell packer isolation integrity
- Flow distribution dynamics

Survey results

Integrated Production Logging and Thermal Modeling Analysis. The comprehensive well evaluation combined multiple diagnostic techniques to assess zonal contributions and completion integrity. [Figure 7](#) presents the integrated results from array production logging tools (PLT) and high-precision temperature measurements alongside 3D thermal modeling outputs.

Data Acquisition and Observations. The temperature profile recorded during production (downpass) appears in blue, while pressure measurements from down and up passes are shown in blue and light blue, respectively. These reveal a gradual pressure transition occurring over the 16-hour survey duration. The 3D model outputs for temperature and pressure (pink dashed line) demonstrate good agreement with measured data.

Flow Profile Analysis. The survey identified active production from 22 distinct zones, with Zones 1, 2, and 5 showing significantly higher contributions than others. The overall flow profile indicates excellent reservoir-wellbore connectivity through the completion screens. A maximum 1% discrepancy between PLT and 3D hydrodynamic temperature model results in Zone 19 likely stems from measurement physics differences and the challenging logging environment near the kick-off point, where wellbore trajectory changes affect flow dynamics and data quality.

Completion Integrity Assessment. Spectral acoustic logging during the uppass focused on evaluating swell packer performance under flowing conditions. Analysis of acoustic signatures revealed:

- The Packers did not show evidence of integrity failure ([Table 3](#)).

- Intact packers displayed consistent acoustic patterns across adjacent zones ([Figure 6](#)).

Table 3—Packer integrity.

Swell packer #	Swell packer integrity
36	Integral
35	Integral
34	Integral
33	Integral
32	Integral
31	Integral
30	Integral
29	Integral
28	Integral
27	Integral
26	Integral
25	Integral
24	Integral
23	Integral
22	Integral
21	Integral
20	Integral
19	Integral
18	Integral
17	Integral
16	Integral
15	Integral
14	Integral
13	Integral

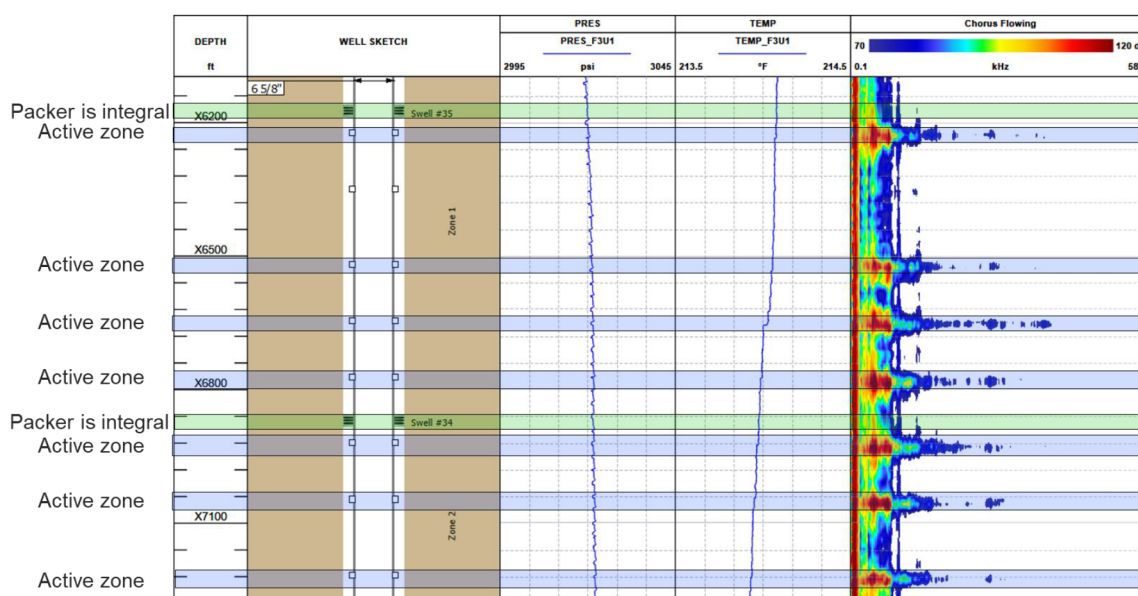


Figure 6—Spectral acoustics shows the integrity of the packer during the flowing regime.

Model Validation. The 3D solver for the thermal model demonstrated strong predictive capability, with the reconstructed oil production profile closely matching PLT measurements. Key findings include:

- Accurate accounting of compartmentalized flow contributions.
- Reliable prediction of inflow distribution along the horizontal section.
- Validated performance in porous media flow conditions.

Conclusions. The integrated approach combining PLT, Acoustic power spectrum, and 3D hydrodynamic temperature modeling provides robust well performance evaluation:

- Thermal modeling shows particular value in challenging logging environments.
- Spectral acoustics offer reliable packer integrity diagnosis.
- The 1% maximum variance between methods falls within acceptable limits for such complex completions.

Table 4 presents the detailed comparison of flow profiles from array PLT and 3D hydrodynamic temperature modeling, demonstrating the consistency between measured and simulated results across all productive zones.

Table 4—Borehole (Array PLT) and Reservoir (3D hydrodynamic temperature modeling) flow distribution in producer #1.

Zone	Borehole Flow (PLT), %	Reservoir Flow (3D hydrodynamic temperature modeling), %
1	6.6	6.4
2	5.6	5.1
3	4.7	4.1
4	3.7	3.7
5	5.1	5.2
6	3.2	3.1

Zone	Borehole Flow (PLT), %	Reservoir Flow (3D hydrodynamic temperature modeling), %
7	4.7	4.2
8	2.9	3.1
9	3	3.3
10	4.4	3.8
11	4.1	4.2
12	2.8	3
13	3.6	3.5
14	3.2	3
15	2.9	2.7
16	2.6	2.8
17	2.5	2.6
18	3.3	3
19	3	4
20	1.7	1.7
21	2.5	3.1
22	1.8	2.6
Below HUD	22.1	21.8

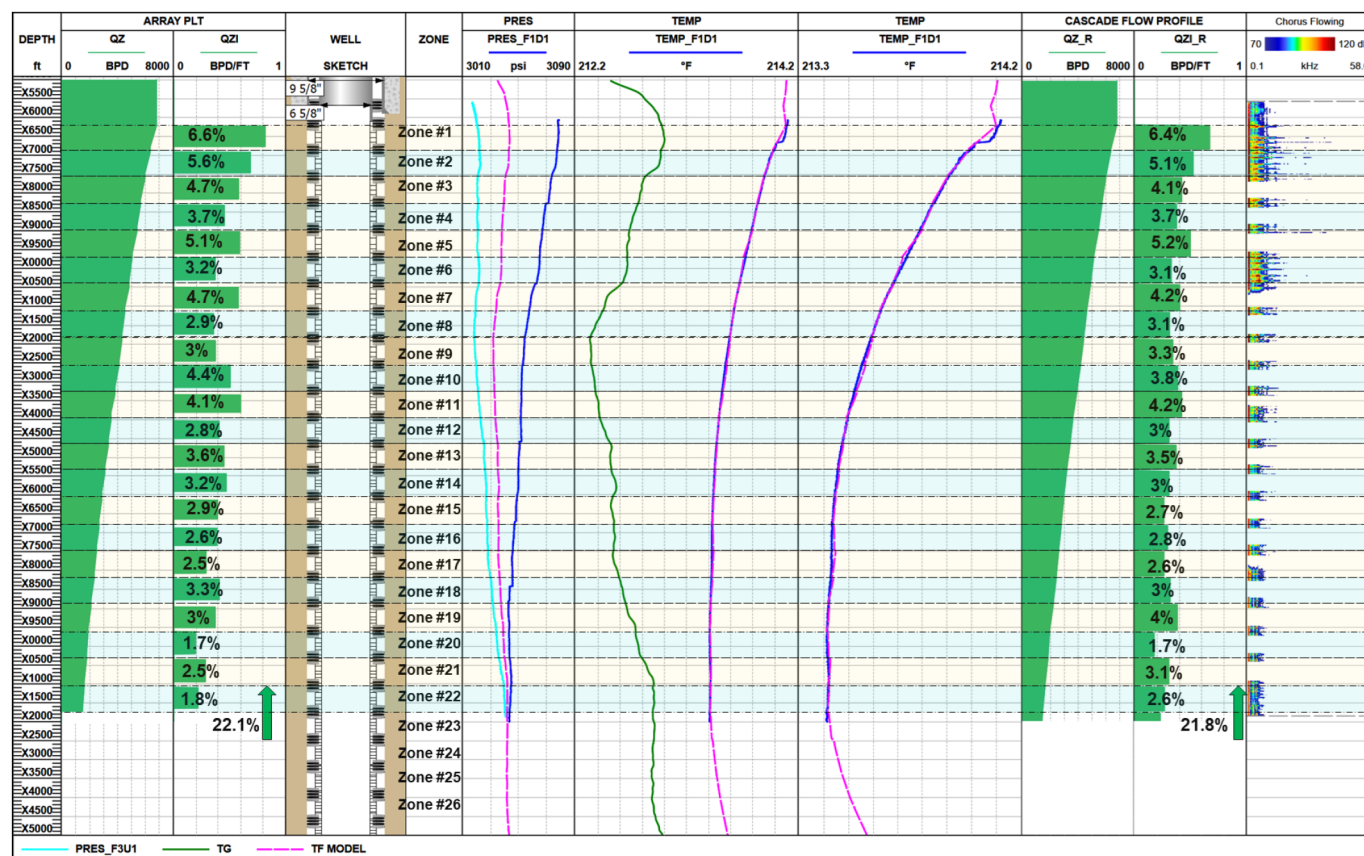


Figure 7—Producer #1 temperature modelling results. Good correlation between temperature modelling and array PLT in most of the zones

Case Study #2

Well history

Extended Reach Well Survey and Analysis. The offshore horizontal producer Well #2 represents an advanced Extended Maximum Reservoir Contact (EMRC) completion in the carbonate formation. Drilled and completed in January 2019, the well includes a 24,753 ft (7.54 km) lateral section — an impressive single-bore horizontal completion for carbonate reservoirs, particularly for its time, though more recent wells have since exceeded this length.

Completion Details. The 6.625" limited-entry liner completion is equipped with swellable packers that segment the lateral into 31 individual production zones. This setup allows for managed inflow distribution along the entire drain length, while preserving the mechanical integrity of the well. Importantly, the well met its production objectives without the need for hydraulic fracturing, highlighting the efficiency of the extended reservoir contact strategy in this formation.

Production History. After 18 months of continuous production, the well maintained a stable flow rate of 9,000 BOPD at the time of the June 2021 survey. The production stream showed minimal water cut or gas breakthrough, confirming effective sweep efficiency across the lengthy completion. The well was not stimulation at the time of the logging operation.

Advanced Diagnostic Survey. The comprehensive June 2021 evaluation employed cutting-edge downhole monitoring technology, High-precision temperature logging, and Acoustic power spectrum tools.

These tools successfully characterized 20 of the 31 completion zones, providing crucial insights into:

- Zonal production contributions
- Swell packer isolation integrity
- Flow distribution dynamics

Survey results

Integrated Production Logging and Thermal Modeling Analysis. The comprehensive well evaluation combined multiple diagnostic techniques to assess zonal contributions and completion integrity. [Figure 10](#) presents the integrated results from array production logging tools (PLT) and high-precision temperature (HPT) measurements alongside 3D thermal modeling outputs.

Data Acquisition and Observations. The temperature profile recorded during production (downpass) appears in blue, while pressure measurements from down and up passes are shown in blue and pink, respectively. These reveal a gradual pressure transition occurring over the 20-hour survey duration. The 3D solver model outputs for temperature (light blue dashed line) and pressure (black dashed line) demonstrate good agreement with measured data.

Flow Profile Analysis. The survey identified active production from 18 distinct zones, with Zones 1, 12, 13, and 14 showing significantly higher contributions than others. The overall flow profile indicates excellent reservoir-wellbore connectivity through the completion screens. A maximum 15% discrepancy between PLT and 3D hydrodynamic temperature modelling results in Zone 1 likely stems from measurement physics differences and the challenging logging environment near the kick-off point, where wellbore trajectory changes affect flow dynamics and data quality.

Completion Integrity Assessment. Spectral acoustic logging during the uppass focused on evaluating swell packer performance under flowing conditions. Analysis of acoustic signatures revealed:

- Three packers showed clear evidence of integrity failure ([Table 5](#)).
- Intact packers displayed consistent acoustic patterns across adjacent zones ([Figure 8](#)).
- Leaking packers generated distinct acoustic anomalies visible on spectral displays ([Figure 9](#)).

Table 5—Packer integrity.

Swell packer #	Swell packer integrity
33	Integral
32	Integral
31	Leaking
30	Integral
29	Leaking
28	Integral
27	Integral
26	Integral
25	Leaking
24	Integral
23	Integral
22	Integral
21	Integral

Swell packer #	Swell packer integrity
20	Integral
19	Integral
18	Integral
17	Integral
16	Integral
15	Integral
14	Integral
13	Integral
12	Integral

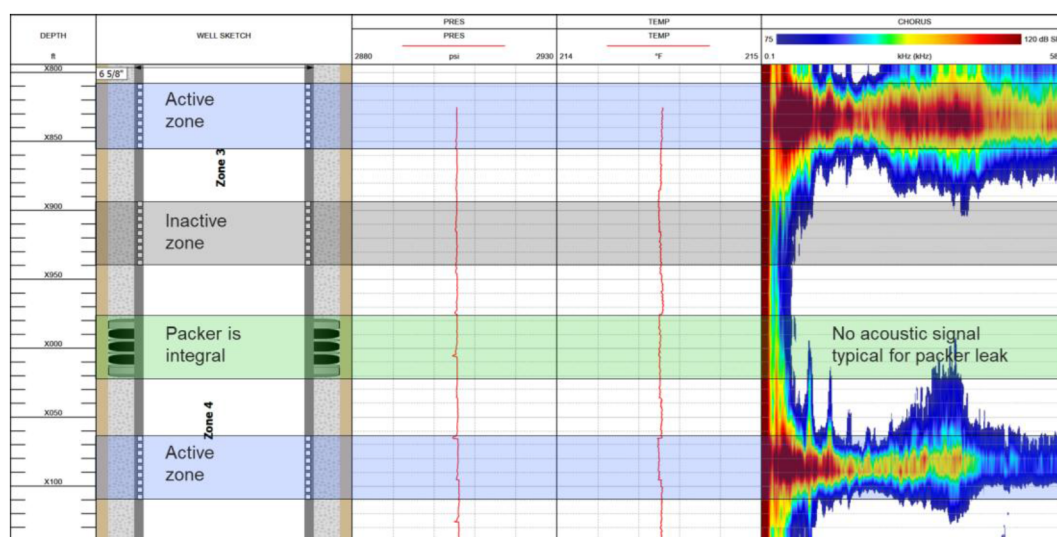


Figure 8—Spectral acoustics shows the integrity of the packer during the flowing regime.

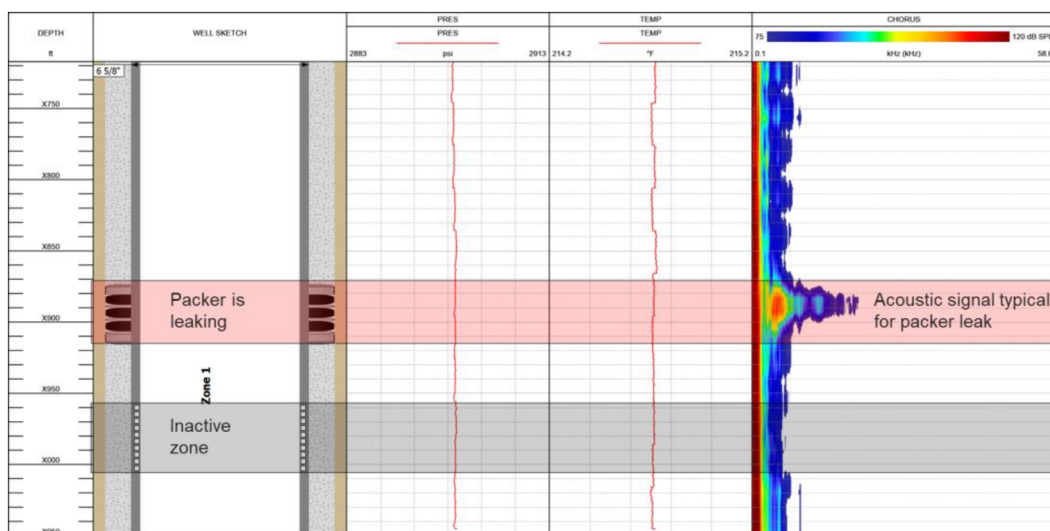


Figure 9—Spectral acoustics indicates a packer leak during the flowing regime.

Model Validation. The 3D thermal model demonstrated strong predictive capability, with the reconstructed oil production profile closely matching PLT measurements. Key findings include:

- Accurate accounting of compartmentalized flow contributions
- Reliable prediction of inflow distribution along the horizontal section
- Validated performance in porous media flow conditions

Conclusions. The integrated approach combining PLT, Acoustic power spectrum, and 3D hydrodynamic temperature modeling provides a robust well performance evaluation.

- Thermal modeling shows particular value in challenging logging environments.
- Spectral acoustics offer reliable packer integrity diagnosis.
- The 15% maximum variance between methods falls within acceptable limits for such complex completions

Table 6 presents the detailed comparison of flow profiles from array PLT and temperature modeling, demonstrating the consistency between measured and simulated results across all productive zones.

Table 6—Borehole (Array PLT) and Reservoir (3D temperature modeling) flow distribution in producer #2.

Zone	Borehole Flow (PLT), %	Reservoir Flow (3D hydrodynamic temperature modeling), %
1	9.8	24.3
2	1.6	1.2
3	5.0	2.3
4	3.2	2.0
5	1.4	0.1
6	5.1	1.5
7	1.1	0.6
8	0.8	0.7
9	4.0	1.4
10	3.6	1.0
11	3.7	0.1
12	6.7	11.0
13	4.7	7.8
14	3.3	4.3
15	1.0	0.3
16	7.2	0.3
17	7.8	3.5
Below HUD	30.0	37.6

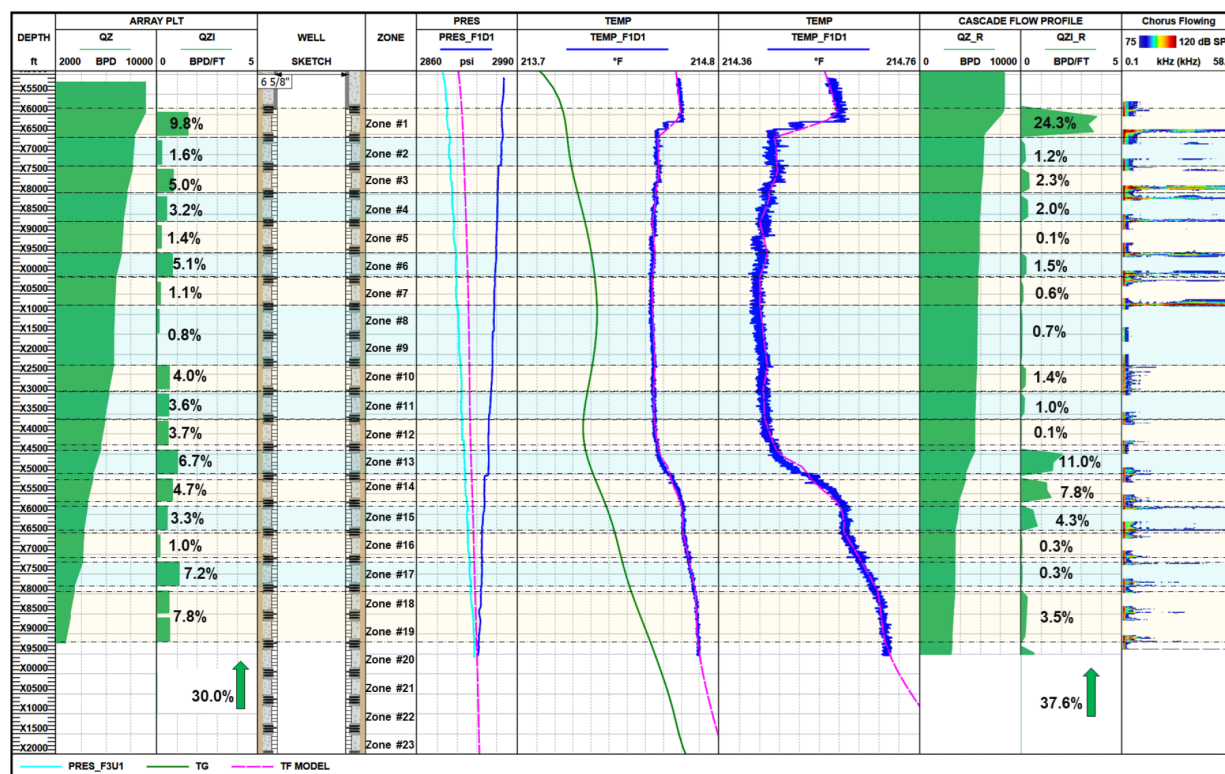


Figure 10—Producer #2 temperature modelling results. Good correlation between temperature modelling and array PLT in most of the zones

Conclusions

This study has successfully validated the effectiveness of an advanced logging methodology combining acoustic power spectrum logging and high-precision temperature measurements, with subsequent 3D hydrodynamic temperature modelling, for accurately determining inflow profiles in horizontal producers. Field applications confirm a high degree of correlation between conventional Production Logging Tool (PLT) results and those derived from High-Resolution Temperature and Acoustic Power Spectrum Logging.

In scenarios where PLT data proves inconclusive - such as low-flow-rate intervals or operational constraints - the proposed temperature-acoustic approach provides a reliable supplementary interpretation, ensuring continuous reservoir performance evaluation.

For operators managing Extended Maximum Reservoir Contact (EMRC) wells, this integrated technique presents a cost-efficient and operationally streamlined alternative to traditional PLT. In spite of premature swelling and significant swell packer debris recovered at the surface while running liner, all swell packers were found integral for Well #1 post stimulation. Operation. Well # 2 was not stimulated at the time of logging, so after diagnostics work, Bull head acid treatment was planned to improve injectivity and allow the non-integral swell packers to swell with exposure to water as the well was flowing < 5% BS&W.

By eliminating the need for complex downhole tool deployments, it reduces both non-productive time (NPT) and associated costs while delivering critical insights into near-wellbore and far-field fluid dynamics. The method's ability to resolve flow contributions from behind the liner - particularly in heterogeneous or fractured formations - enhances reservoir management decision-making, enabling timely interventions to mitigate water or gas breakthrough. The current results underscore the viability of this approach as a next-generation surveillance tool for complex well architectures.

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References

Journals

- Alford, M. C. and Bennett, R. G., 2006, Distributed Temperature Sensing in Oil and Gas Well Applications, *SPE Journal*.
Cinco, L.H., Samaniego, V.F., Domingues, N., Transient pressure behavior for a well with a finite-conductivity vertical fracture, 1978, *SPE Journal*, **18**(4), 253–264, DOI: [10.2118/6014-PA](https://doi.org/10.2118/6014-PA).
Khokhar, M.M. and Economides, M.J., 2006, Production Logging: An Essential Tool for Optimal Reservoir Management, *Journal of Petroleum Science and Engineering*.
Khokhar, M.M. and Economides, M.J., 2000, Production Logging: Theoretical and Practical Aspects, *Journal of Petroleum Science and Engineering*.
Ouyang, L.B., Arbabi, S. and Aziz, K., 1998, A single-phase wellbore-flow model for horizontal, vertical, and slanted wells, *SPE Journal*, **3**(2), 124–133. DOI: [10.2118/36608-PA](https://doi.org/10.2118/36608-PA)
Ozkan, E., Sarica, C., Haciislamoglu, M., and Raghavan, R., 1993, Influence of pressure drop along the wellbore on horizontal-well productivity, *SPE Journal*, **4**(3), 288–301, DOI: [SPE-25502-MS](https://doi.org/10.2118/SPE-25502-MS)

Conference Papers

- Aslanyan, A., Aslanyan, I., Salamatina, A., Karuzin, A., Fesina, Yu., Zaripov, I., Skutin, V., AL Ghafri, Z., AL Yarabi, M., and AL Maharbi, A., 2014, Numerical Temperature Modelling for Quantitative Analysis of Low- Compressible Fluid Production, Paper SPE-172090-MS presented at The Abu Dhabi International Petroleum Exhibition and Conference, Abu-Dhabi, UAE, 10-13 November 2014. DOI: [10.2118/172090-MS](https://doi.org/10.2118/172090-MS).
Aslanyan, A.M., Wilson, M., Al-Shammakhy, A., and Aristov, A., 2013, Evaluating Injection Performance with High-Precision Temperature Logging and Numerical Temperature Modelling, Paper SPE 166007-MS presented the SPE Reservoir Characterization and Simulation Conference and Exhibition, Abu Dhabi, UAE, September 2013. DOI: [10.2118/166007-MS](https://doi.org/10.2118/166007-MS)
Virt, V., Kosolapov, V., Nagimov, V., Salamatina, A., Fesina, Yu., Alekseeva, A., Yakhina, Yu., and Skutina, E., 2021, Individual Fracture Efficiency Monitoring In Horizontal Wells By Using A New 3D Fine-grid Temperature Modelling, Paper SPE-207237-MS presented at The Abu Dhabi International Petroleum Exhibition and Conference, Abu-Dhabi, UAE, 15-18 November 2021. DOI: [10.2118/207237-MS](https://doi.org/10.2118/207237-MS).
Volkov, M., Nagimov, V., Avilez, S., Lazutkin, D., Khachatryan, M., and Prodan, A., 2022, Unconventional reservoir flow assessment using Spectral acoustics and numerical temperature modelling, Paper URTEC-3721549-MS presented at the SPE/AAPG/SEG Unconventional Resources Technology Conference, Houston, Texas, USA, 20-22 June 2022. DOI: [10.15530/urtec-2022-3721549](https://doi.org/10.15530/urtec-2022-3721549)

Meeting Paper Included in a Printed Proceedings Volume

- Alekseeva, A., Salamatina, A., Fesina, Yu, Trusov, A., Kononov, S., Pimantsev, V., 2022, Flow modelling in horizontal production wells with hydraulic fractures, The 5th Anniversary EAGE Scientific and Practical Conference Horizontal Wells 2022, Sochi, Russia, 10-14 October 2022.

Books, PhD Theses, and Reports

- Altshul, A.D., 1970, *Gidravlicheskie soprotivleniia, Nedra, Moscow*. [Altshul, A.D., Hydraulic resistance, Publishing house Nedra, Moscow]
Bratland, O., 2010, *Pipe Flow 2: Multiphase Flow Assurance*. ISBN: 978-616-335-926-1.
Carslaw, H.S., Jaeger, J.C., 1959, *Conduction of Heat in Solids*, Clarendon Press, Oxford. ISBN: 0198533683
Charny, I.A., 1961, *Osnovy gazovoi dinamiki, Gosudarstvennoe nauchno-tekhnicheskoe izdanie neftianoi i gornotoplivnoi literatury, Moscow*. [Charny, I.A., 1961, *Basics of gas dynamics*, State Scientific and Technical Publishing House of Oil and Mining and Fuel Literature, Moscow]

- Charnyj, I.A., 1963, *Podzemnaya gidrogazodinamika*, Gosudarstvennoe nauchno-tekhnicheskoe izdatel'stvo neftyanoj i gorno-toplivnoj literatury, Moscow. [Charnyj, I.A., 1963, *Underground hydrodynamics*, State Scientific and Technical Publishing House of Oil and Mining and Fuel Literature, Moscow.] ISBN: 5-93972-591-0.
- Chekaliuk, E.B., 1965, *Termodinamika neftyanogo plasta*, Nedra, Moscow. [Chekalyuk, E.B., 1965, *Thermodynamics of an oil reservoir*, Publishing house Nedra, Moscow]. ISBN: 978-5-458-36108-8.
- Dzhaluriya, Yo, 1980, *Estestvennaia konvektciya*, Mir, Moscow. [Transl.: Jaluria, Yo, 1980, *Natural convection*, Oxford]
- Keis, V.M., 1972, *Konvektivnyi teplo- i massoobmen*, Energiya, Moscow. [Transl.: Kays, W.M., 1972, *Convective heat and mass transfer*, Stanford University, Mc Graw-Hill Book Company]
- Mikheev, M.A., Mikheeva, I.M., 1973, *Osnovy teploperedachi*, Energiya, Moscow. [Mikheev, M.A., Mikheeva, I.M., 1973, *Basics of heat transfer*, Energiya, Moscow]. ISBN: 978-5-903178-20-9
- Paskonov, V.M., Polezhaev, V.I., and Chudov, L.A., 1984, *Chislennoe modelirovanie protsessov teplo- i massoperenosa*, Nauka, Moscow, *Glavnaia redaktsiia fiziko-matematicheskoi literatury*. [Paskonov, V.M., Polezhaev, V.I., and Chudov, L.A., 1984, *Numerical modelling of heat and mass transfer processes*, Nauka, Moscow, Main edition of physical and mathematical literature].
- Pudovkin, M.A., Salamatina, A.N., and Chugunov, V.A., 1977 *Temperaturnye protsessy v deistviushchikh skvazhinakh*, Izdatel'stvo Kazanskogo universiteta, Kazan. [Pudovkin, M.A., Salamatina, A.N., and Chugunov, V.A., 1977, *Temperature processes in operating wells*, Publishing house of Kazan University, Kazan]
- Tihonov, A.N., Samarskii, A.A., 1966, *Uravneniia matematicheskoi fiziki*, Nauka, *Glavnaya redaktsiia fiziko-matematicheskoi literatury*, Moscow. [Tihonov, A.N., Samarskii, A.A., 1966, *Equations of mathematical physics*, Nauka, Main edition of physical and mathematical literature, Moscow]. ISBN: 5-211-04843-1.
- Samarskii, A.A., 1971, *Vvedenie v teoriyu raznostnykh skhem*, Nauka, Moscow, *Glavnaia redaktsiia fiziko-matematicheskoi literatury*. [Samarskii, A.A., 1971, *Introduction to the theory of difference schemes*, Nauka, Moscow, Main edition of physical and mathematical literature]. ISBN 5-02-014576-9